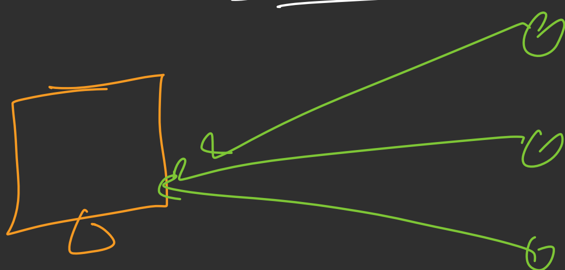
Introduction to DevOps

HTML + CSS + JS

↳ static websites

Developer



Server

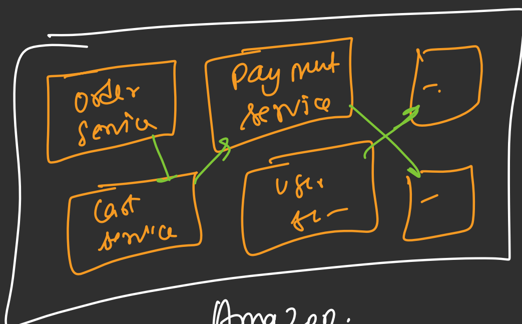
Idea

↳ Coding

↳ Deploy

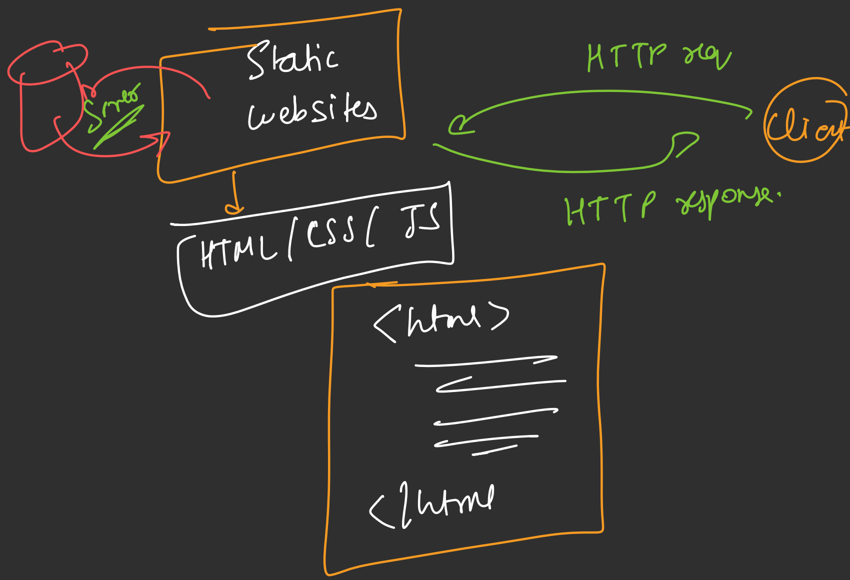
Netflix
Amazon
X
Instagram

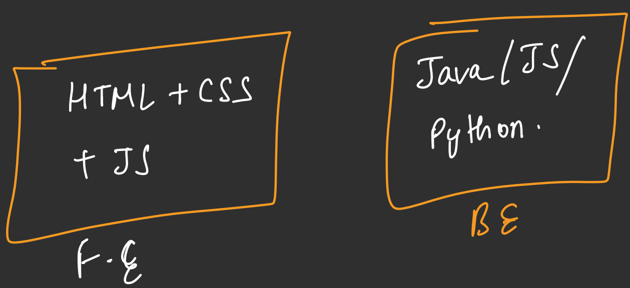
- Auth / Authorization.
- Data intensive.
- Multiple server
- Microservice

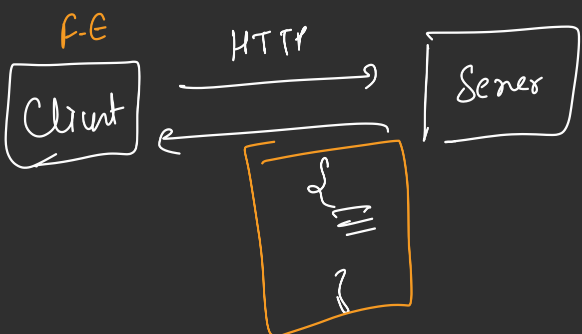


Amazon.

Operations


Static websites
Server
HTTP req
Client
HTTP response.
HTML / CSS / JS
<html>
</html


HTML + CSS + JS
F.E
Java / JS / Python.
BE


F-E
HTTP
Client
Server
{
}

Two World

Developers

Code
→ new Features
→ Bug Fixes

Operations

Deploy
→ Multiple servers.
→ DB Mange
→ Config
→ OS

Java 21

Testing

Dev → System Change

Ops → Not to Change

DevOPS

Developers

Operations.

DevOps

- Role
- $\boxed{\text{Philosophy}}$

Fundamental ideas

1. Shared Ownership.
2. Automation.
3. Small & Regular
4. Fast feedback

DevOps philosophy

Dev

Operation

Silo working Model

DevOps

Devops

Fast Delivery

Reliable Delivery.

CALMS Framework

- C → Culture
- A → Automation
- L → Lean
- M → Measurement
- S → Sharing

Dev	QA	Ops
↓		
5 days	20d	20d

DORA Matrix

⇓

Devops Research And Assesment

→ Deployment freq
→ Lead Time for change
→ Mean Recovery Time
→ Change Failure Ratio

Q) How freq do you deploy?

Q) How quickly does a change reach to prod?

Q) How freq failure occurs in deployment?

Q) How Quickly do you recover?

Deployment freq

- per day
- per week
- per Month

Deployment

↳ Patch
↳ Bug fix
↳ feature.

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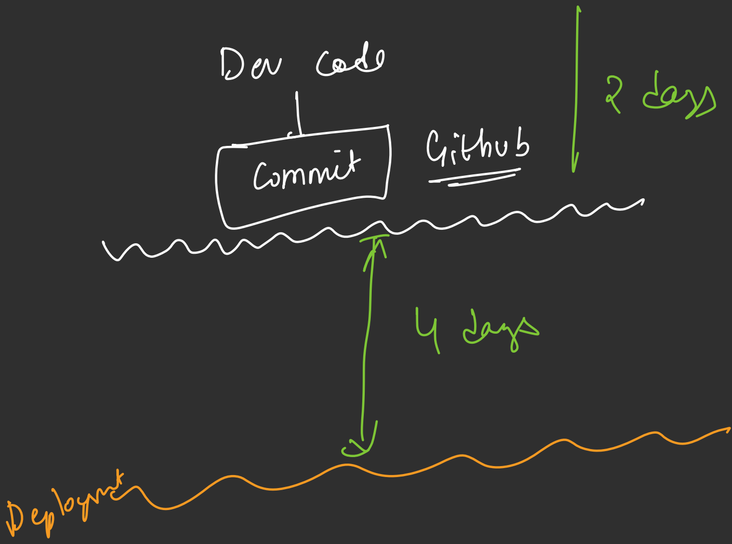
40 product

20 days

$$= \frac{40}{20} = 2 \text{ deployment/day.}$$

② Lead time for change.



- Change Failure Rate.

$$\boxed{= \frac{\text{Failed Deployment}}{\text{Total deployment}} \times 100}$$

- Failure Recovery Time.

F1 : 20 M

F2 : 40 M

F3 : 90 M

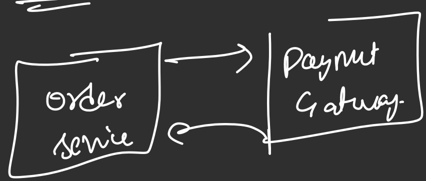
$$= \frac{20 + 40 + 90}{3}$$

$$= \frac{150}{3} = \boxed{50 \text{ Min}}$$

Recovery

- fix. (hotfix)
- Revert
- Disable

Dev



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	SUPER200	T

Prod Env

Static

↓

Complex websites

(Dev + ops)

↓

SILO → (1) fast delivery
(2) Reliable

↓

DevOps

↓

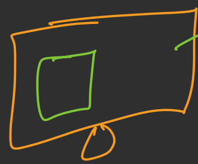
CALMS

↓

DORA

MAC

↳ VM

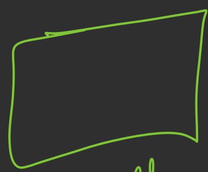
 → linux

Computer (server)

UBUNTU

↳ VM

↳ WSL



Shell

LINUX

Operating System

User → OS → HW

↳ CPU

↳ RAM

↳ Memory (Disk)

↳ Keyboard

↳ N/W.

Kernel

↳ Linux

↓

memory

process (Program).

N/w

user permission.

system call.

filesystem.

Java / JS   File

↓

OS API

↓

System calls (Kernel)

↓

Data Return

User space

Kernel space.

Computer

BUG

Files
programs

Java/JS

System.out.println ()

cout < <

(System call)

OS

LINUX → Distribution

(Kernel)

Command line tool

Shell

Service Manager

Ubuntu

Kali

Fedora

Debian

Ubuntu / MAC

GUI

cd Desktop

ls

Cd Desktop

Terminal

Program

(user space).

Shell

Bash

Zsh → MAC OS

sh

Fish

cd /home/Desktop.

>>

Terminal

Shell (Bash / Zsh)

cd home/Desktop

Kernel

cd

tmp